MECH559: Term Project Prproposal

Optimization of Log-Yard Operations Using Discrete-Event Simulation

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Abstract

This project will investigate the optimization of forestry mill-yard operations using a discrete-event simulation. The simulated yard will contain stochastic truck arrivals, multiple log species, finite-capacity log piles, mobile loaders, a mill-infeed buffer, and variable downstream demand. These elements compete for shared handling capacity: incoming trucks must be unloaded while the mill must be continuously supplied with the appropriate log species. The central design problem is to determine a state-dependent loader-dispatch policy that coordinates these competing requests. Candidate policies will be evaluated using repeated stochastic simulations and compared using measures such as throughput, truck waiting time, mill-starvation time, loader travel, energy use, and inventory shortage. The project will combine subsystem-level optimization with system-level gradient-free optimization methods to handle the nonsmooth model.

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1 Proposed Project and Overview

The proposed project is entitled *Optimization of Log-Yard Operations Using Discrete-Event Simulation*. A forestry mill yard coordinates the arrival, storage, handling, and processing of large quantities of logs. Although the physical operations may appear straightforward, the yard is a coupled system with several competing demands. Trucks arrive at uncertain times and carry different species or grades of logs. Loaders must unload trucks, move material between piles, and replenish the mill infeed. The mill has time-varying material requirements and may be penalized heavily when the required log type is unavailable.

The project will develop a simplified but useful model of this system. The model will support stochastic and prescribed truck arrivals, multiple log species, finite-capacity log piles, mobile loaders, a mill-infeed buffer, and a simplified representation of downstream demand. Material handling will be represented as a series of possible loader tasks. For example, a loader may unload an arriving truck into a pile, move material between piles, replenish the mill with spruce from a pile, replenish the mill with pine from a newly unloaded truck, or remain idle. The operating policy must determine which task should be performed by which loader and when.

The central research question is:

> How should mobile loaders be allocated among truck unloading, pile management, and mill-infeed replenishment so that the yard achieves high throughput and reliable mill operation while limiting truck waiting, equipment movement, energy use, and inventory imbalance?

The project will both characterize the performance of a simulated mill yard and select configuration and operating rules that achieve desirable system-level performance. The design variables may include physical decisions, such as the number of loaders and buffer capacities, as well as operational decisions, such as task priorities, inventory thresholds, and loader-dispatch rules.

2 Proposed Methodology

The methodology will be developed in stages. First, a sufficiently simple system model will be defined and baseline operating policies will be established and checked against expected behaviour. Next, each subsystem will be investigated through its associated design variables and objectives. The subsystems will then be integrated into a common discrete-event simulation. Finally, the integrated system will be optimized and evaluated under multiple operating scenarios.

The exact subsystem boundaries may be revised during model development. This is intentional: the initial decomposition is a working hypothesis, and it may be determined that some elements should be combined or that another element deserves independent treatment once the interactions and computational effort become clearer.

2.1 The Optimization Problem

At a given simulation time, the condition of the yard can be described by a state vector, $\boldsymbol{x}_t$, such as

$$\boldsymbol{x}_t = \left[\boldsymbol{x}_t^{\text{truck}}, \boldsymbol{x}_t^{\text{pile}}, \boldsymbol{x}_t^{\text{infeed}}, \boldsymbol{x}_t^{\text{loader}}, \boldsymbol{x}_t^{\text{mill}}, \boldsymbol{x}_t^{\text{demand}}\right]. \tag{1}$$

The state may include the trucks waiting to be served, their waiting times, species, and volumes; the inventory and capacity of each pile; the material currently in the mill infeed; the location and availability of each loader; the current mill-processing state; and current or anticipated demand. The state changes when discrete events occur, such as truck arrivals, loader-task completion, material transfer, demand changes, or mill starvation.

At each decision instance, the mill operator selects a loader action from the feasible task set $\mathcal{T}(\boldsymbol{x}_t)$. Decision instances occur when an actionable change has taken place, such as when a loader becomes available or when a new truck or urgent mill request appears. A task may involve unloading a specific truck or transferring a specified log species from a pile to the mill infeed. Drawing on the notation of control theory, the mapping from state to action can be written as

$$\boldsymbol{u}_t = \boldsymbol{\pi}(\boldsymbol{x}_t; \boldsymbol{\theta}), \tag{2}$$

where $\boldsymbol{u}_t$ is the loader action, $\boldsymbol{\pi}$ is a state-dependent decision policy, and $\boldsymbol{\theta}$ contains the policy parameters. In the initial policy, these parameters will primarily be weights that express the relative importance of truck waiting, mill-starvation risk, inventory shortage, and loader travel distance. Thresholds and targets may later be added as policy parameters if they are needed to calculate the features or define task urgency.

For each feasible task, a state-dependent priority function will be calculated and combined linearly with the policy parameters to produce a total priority score:

$$P_j(\boldsymbol{x}_t; \boldsymbol{\theta}) = \boldsymbol{\theta}^{\mathsf{T}} \boldsymbol{\phi}_j(\boldsymbol{x}_t), \tag{3}$$

where $\boldsymbol{\phi}_j$ is a task-specific feature vector for task j, for example:

$$\boldsymbol{\phi}_j(\boldsymbol{x}_t) = \begin{bmatrix} \text{truck waiting age} \\ \text{mill starvation risk} \\ \text{species shortage severity} \\ \text{loader travel time} \\ \text{infeed deficit} \\ \text{task-type indicator} \\ \vdots \end{bmatrix}. \tag{4}$$

Each task-specific feature vector contains the selected factors relevant to assessing that task's urgency and handling cost. A feature that does not apply to a particular task may be set to zero.

For example, truck waiting age is relevant to an unloading task but not to a mill-replenishment task. Infeasible tasks, such as moving spruce from an empty pile, are excluded from the feasible task set before their priorities are compared.

The policy selects the task with the highest priority:

$$j_t^\star = \arg \max_{j \in \mathcal{T}(\boldsymbol{x}_t)} \boldsymbol{\theta}^\mathsf{T} \boldsymbol{\phi}_j(\boldsymbol{x}_t), \tag{5}$$

and $\boldsymbol{u}_t$ is the loader action associated with the selected index $j_t^\star$. For the first model, available loaders can be assigned tasks sequentially in priority order. If needed, the action can later be extended to a joint assignment of multiple loaders to multiple tasks.

The parameter vector and the state-dependent priority scores determine the sequence of decisions, and the simulation progresses through a trajectory:

$$\boldsymbol{x}_0 \to \boldsymbol{x}_1 \to \cdots \to \boldsymbol{x}_T. \tag{6}$$

The objective is not calculated from the priority scores themselves. Instead, performance is calculated from the aggregate consequences of the decisions over the simulated horizon T. With stochasticity and disturbances in the model, such as uncertain truck arrivals, truck composition, service times, and demand, the performance of each candidate policy is approximated over several simulation replications:

$$\widehat{\boldsymbol{J}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{k=1}^{N} \boldsymbol{J}(\boldsymbol{\theta}; \boldsymbol{\xi}_k), \tag{7}$$

where $\boldsymbol{\xi}_k$ represents one realization of the uncertain conditions and $\boldsymbol{J}$ contains the performance measures of interest. A gradient-free optimization method will search for a high-performing parameter vector $\boldsymbol{\theta}$, or a set of Pareto-efficient parameter vectors associated with different tradeoffs among the objectives.

The project will not attempt to define an "optimal" action for every mathematically possible state. Instead, it will seek an interpretable operating approach through an optimized parameter vector and an explicitly defined set of state features.

2.1.1 Objectives and Design Tradeoffs

The yard has no single universally correct definition of good performance. The most important objectives are expected to include:

- maximizing processed log volume or mill throughput;
- minimizing mean and high-percentile truck waiting time;
- minimizing mill-starvation time and interruptions;
- minimizing loader travel, handling, and energy use;

- reducing inventory shortages and species imbalance; and
- maintaining acceptable loader utilization and operational robustness.

These objectives conflict. Prioritizing truck unloading can reduce truck queues but leave the mill without the required species. Prioritizing mill replenishment can preserve production continuity but increase truck waiting. Increasing safety stock can reduce the probability of mill starvation but requires more storage space and may increase handling. Adding loaders can improve responsiveness but may increase capital cost, energy use, interference, and idle capacity during low-demand periods. Similarly, operating all resources near full utilization may appear efficient but can make the system sensitive to small arrival delays or demand surges.

The project will initially retain the objectives separately and use Pareto analysis to show the tradeoffs between them. A normalized weighted objective could later be used to select a single preferred operating point:

$$J_{\boldsymbol{\lambda}}(\boldsymbol{\theta}) = \boldsymbol{\lambda}^{\mathsf{T}} \widehat{\boldsymbol{J}}(\boldsymbol{\theta}), \tag{8}$$

where $\boldsymbol{\lambda}$ represents selected preferences among normalized objectives. Quantification of very undesirable events, such as 95th-percentile truck waiting time and prolonged mill interruption, may also be included, since averages alone can hide poor rare-event performance.

2.1.2 Expressiveness and Limitations of the Parameterized Policy

Both the state-dependent features and the parameter vector drive the decision process. For example, a mill-replenishment task may receive a score of the form

$$P_{\text{mill}} = \theta_M \left(\frac{I_{\text{target}} - I_{\text{infeed}}}{I_{\text{target}}} \right) + \theta_S \cdot \text{starvation-risk}, \tag{9}$$

where I is an infeed-volume measure, while a truck-unloading task may receive

$$P_{\text{truck}} = \theta_T \cdot \text{truck waiting time} + \theta_Q \cdot \text{truck queue length}. \tag{10}$$

Early in a shift, the infeed may be full and truck urgency may dominate:

$$P_{\text{truck}} > P_{\text{mill}}. \tag{11}$$

Later, as the mill consumes spruce and the infeed approaches empty, starvation risk rises:

$$P_{\text{mill}} > P_{\text{truck}}. \tag{12}$$

The same policy parameters can therefore switch naturally from truck-first to mill-first as the state changes. In this sense, the optimizer is choosing the relative importance of the factors and the conditions under which the dispatching tendency changes.

Linear priority scoring is convenient because it is transparent, straightforward to implement, and allows the reason for a particular decision to be inspected. It may not capture every useful interaction in the operating process. If it proves too limited, the policy can be made more expressive without changing the overall optimization structure. Possible extensions include:

- nonlinear features such as (truck age)^2, so old trucks become disproportionately urgent;
- threshold features, for example making mill urgency very high when the infeed is below a specified volume;
- remaining-horizon time, to account for the end of a shift;
- demand forecasts or predicted truck arrivals; and
- different parameters for different species or loader locations.

Over specific priorities also have the risk of guiding the optimization to an assume optimal space and limiting interesting results. Other policy structures can also be considered if they become useful during model development.

2.2 System and Subsystem Definition

The general flow of material through the system is shown conceptually below:

$$\text{truck arrivals} \rightarrow \text{log-pile inventory} \rightarrow \text{mill infeed} \rightarrow \text{mill demand.}$$

The loaders handle the movement between these stages. The log-yard model will initially include the following elements:

- trucks arriving with different species, grades, and volumes;
- a truck queue and one or more unloading locations;
- finite-capacity piles with species-dependent inventories;
- a fleet of mobile loaders with locations, availability, and handling times;
- one or more mill-infeed buffers;
- a simplified mill process and downstream-demand model; and
- dispatch and replenishment priority rules that coordinate the shared resources.

A four-subsystem decomposition is proposed in Table 1. The subsystems focus on defining the priorities of tasks and the capacity available to complete them. Since the group has three members and four subsystems, one member will take responsibility for two of the smaller, closely related

subsystems. The exact allocation of member responsibilities will be confirmed once the model interfaces are finalized.

Subsystem	**Primary design questions**
Inbound trucks and unloading	How should truck arrivals, queue priorities, and unloading assignments be represented and defined for optimization?
Loader fleet	How should handling capacity be selected to satisfy task demand from the other subsystems?
Log-pile inventory and storage	How should pile capacities, species allocation, safety stock, and replenishment thresholds be selected to balance shortages against storage and handling costs?
Mill infeed and downstream demand	How should infeed-buffer targets, species requirements, demand variation, and mill-starvation penalties be represented and coordinated with loader decisions?

Table 1: Tentative subsystem decomposition.

The subsystems are coupled. The inbound subsystem determines when and what material becomes available. The inventory subsystem determines whether material can be accessed when needed. The loader subsystem controls the shared handling capacity. The mill-infeed subsystem creates the downstream urgency that drives replenishment requests. A subsystem-level optimum may therefore be poor at the system level. For example, minimizing truck waiting in isolation could starve the mill, while minimizing mill-starvation time in isolation could create unacceptable truck queues. The simulation and Pareto analysis will be used to quantify and visualize these interactions.

2.3 Simulation Model

The simulation will be implemented in Python using SimPy, a process-based discrete-event simulation package. SimPy provides simulation backend with a virtual clock and event queue. User-defined processes advance by yielding events such as timeouts, resource requests, material transfers, or completion signals. The environment then advances to the next scheduled event rather than stepping through every instant of time [1].

The following objects are anticipated in the log-yard model:

- **Environment:** maintains simulation time and schedules future events over an operating shift.
- **Truck-arrival process:** generates trucks according to a prescribed schedule or a stochastic arrival model. Each truck will have attributes such as arrival time, species composition, volume, and unloading requirements.
- **Loader processes and resources:** represent loader availability, travel, loading, unloading,

location, service time, and possibly energy use. A simple early model may use a shared loader capacity; a more detailed model will represent individual loaders and their locations.

- **Inventory states or containers:** represent the quantities of each species in log piles and in the mill infeed. Capacity constraints and material shortages will be recorded explicitly.
- **Mill process:** consumes the required species from the infeed according to a prescribed or stochastic demand profile. If the required material is unavailable, the resulting waiting period will contribute to mill-starvation metrics.
- **Task queues:** contain pending requests such as unloading a truck or moving a particular species to the mill. The dispatch policy will determine which feasible task is assigned when a loader becomes available.
- **Metrics and event log:** record truck waiting, queue lengths, inventory trajectories, loader busy time and travel, processed volume, starvation time, and the sequence of operational events.

The initial model will deliberately use simplified process-time and travel-time assumptions. Complexity will be added only when it improves the engineering conclusions. Potential extensions include time-varying demand, loader breakdowns, mill breakdowns, service-time distributions, heterogeneous loader capabilities, multiple unloading locations, and pile-specific travel distances. The project will avoid detailed log-level physics, full crane kinematics, and detailed kiln thermodynamics unless these are shown to be necessary for the selected system-level questions.

The simulation will be structured so that the physical model, dispatch policy, and optimization routine are separate. Conceptually, the optimization routine will call

```
metrics = run_simulation(policy_parameters, scenario)
```

The policy parameters will be changed by the outer optimization algorithm, and the simulation will output the information needed to calculate performance. This separation will keep the code base modular: comparing different dispatch policies and optimization methods will not require rebuilding the underlying yard model.

2.4 Optimization Framework

The simulation creates a challenging optimization landscape. Even if the truck schedule and demand are fixed, the problem is generally not smooth. A small change in policy parameters can change which of several tasks is selected first. That change can alter the entire subsequent event sequence: loader locations change, queues develop differently, material reaches the mill at different times, and later decisions will almost certainly be different. The resulting objective is therefore nonsmooth with respect to policy parameters. When arrivals and demand are stochastic, the performance estimate also contains noise.

For these reasons, conventional gradient-based optimization is not expected to be reliable as the primary method for this system. The project will use gradient-free optimization methods. The final method will be selected based on the model and the number and type of design variables in the problem. Candidate methods include grid or random search, local perturbation search, evolutionary methods, and Bayesian methods. In general, these methods evaluate candidate designs using their simulated performance, smartly search the design space isolating more feasible area, use understanding of uncertainty to motivate exploration, or locally approximating the design space as linear [2].

Each candidate policy will be evaluated on multiple random scenarios. During optimization, the same scenarios may be used for competing policies so that differences are not dominated by noise. Final validation will use independent scenarios that were not used to select the policy. The optimized policies will also be compared with simple baselines, including truck-first, mill-first, oldest-request-first, and other logical heuristics.

The intended output is not necessarily one universally optimal policy. A more useful result may be a set of policies representing different operating preferences:

- a high-throughput policy that accepts longer truck waits;
- a truck-service policy that minimizes inbound delays;
- a low-energy policy that limits travel and unnecessary handling;
- a balanced policy with acceptable performance across all objectives; and
- a robust policy that limits high-percentile delays and mill interruptions.

3 Anticipated Modeling Effort and Project Scope

The model will be developed in stages to avoid an unnecessarily detailed simulation. The initial minimum viable model will contain one shift, two or three loaders, a truck queue, two or three log species, finite-capacity piles, one mill-infeed buffer, stochastic or prescribed truck arrivals, and stochastic or prescribed demand.

The anticipated stages are:

1. Define the state variables, events, subsystem interfaces, and performance metrics.
2. Implement and verify a simple deterministic simulation with fixed arrivals and service times.
3. Add stochastic arrivals, species composition, demand, and process times.
4. Implement baseline dispatch policies and verify that the simulation reproduces expected qualitative behaviour.
5. Add subsystem-level design variables and optimization experiments.

6. Integrate the subsystems and perform multiobjective system-level optimization.

7. Validate selected policies on independent scenarios and report sensitivity to uncertain assumptions.

The simulation will be verified by checking conservation of material, resource availability, queue behaviour, event ordering, and limiting cases. For example, the model should show no mill starvation when infeed supply is unlimited, and truck waiting should decrease when unloading capacity is increased while all other conditions are fixed. The model's limitations and assumptions will be further explored throughout the project stages and thoroughly discussed in the final report.

4 Expected Outcome

The project will produce a computational framework for studying mill-yard operations and a quantitative comparison of alternative loader-dispatch strategies. The main expected outcome is a demonstration of how a coupled, discrete, stochastic process system can be decomposed into interacting subsystems, modeled using discrete-event simulation, and optimized using derivative-free methods. The results should also clarify which operational tradeoffs are most important: truck service versus mill continuity, inventory robustness versus storage and handling, and throughput versus energy and resilience.

References

[1] SimPy Development Team, *SimPy Documentation*, `https://simpy.readthedocs.io/`, accessed September 2026.

[2] J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed., Springer, 2006.